EXP 11 a) SIMULATE STATIC ROUTING CONFIGURATION USING CISCO PACKET TRACER.

AIM:- Simulate static routing configuration using Cisco Packet tracer.

PROCEDURE :-

Setting up a practice lab.

Creating, adding, verifying static routes connected network of each router

Router	Available on local	Available on other router
Router 0	10.0.0.0/8 20.0.0.0/8	30.0.0.0/8, 50.0.0.0/8.
Router 1	10.0.0.0/8, 30.0.0.0/8	10,0.0.0/8 40.0.0.0/8

Router 0 configuration :-

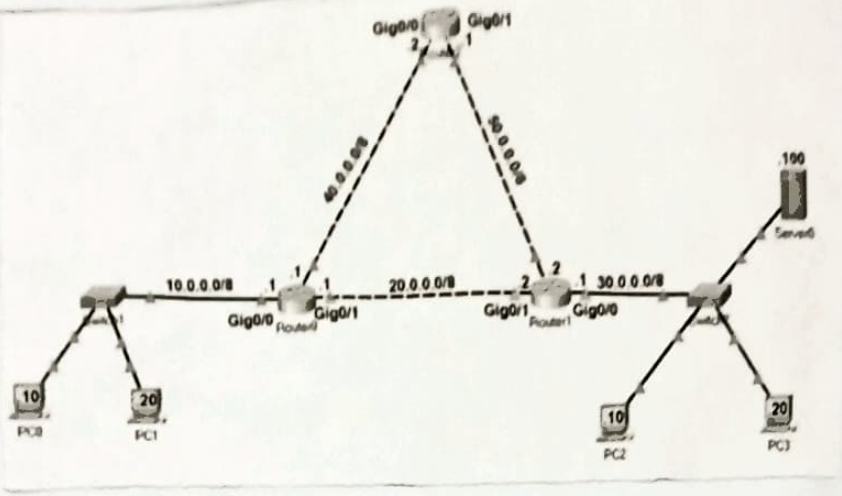
Router 0 enable

Router # configure terminal

Router (config) #ip router 30.0.0.0 255.0.0.0 20.0.0.2

Router (config) #ip route 30.0.0.0 255.0.0.0 40.0.0.200

Router (config) #exit



Router 1. configuration

Router > enable

Router # configuration terminal

Router (config) # ip route 10.0.0.0 255.0.0.0 20.0.0.1 10.

Router (config) # ip route 10.0.0.0 255.0.0.0 50.0.0.1 20.

Router (config) # exit

Router0

Physical Config CLI Attributes

IOS Command Line Interface

```
Router>enable
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#ip route 30.0.0.0 255.0.0.0 20.0.0.2 10 Primary route
Router(config)#ip route 30.0.0.0 255.0.0.0 40.0.0.2 20 Backup route
Router(config)#ip route 30.0.0.100 255.255.255.255 40.0.0.2 10 Primary route
Router(config)#ip route 30.0.0.100 255.255.255.255 20.0.0.2 20 Backup route
Router(config)#ip route 50.0.0.0 255.0.0.0 40.0.0.2 10 Primary route
Router(config)#ip route 50.0.0.0 255.0.0.0 20.0.0.2 20 Backup route
Router(config)#exit
Router#show ip route static
     30.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
S       30.0.0.0/8 [10/0] via 20.0.0.2
S       30.0.0.100/32 [10/0] via 40.0.0.2
S    50.0.0.0/8 [10/0] via 40.0.0.2

Router#
```

Router adds only primary routes to the routing table.

verify static routing

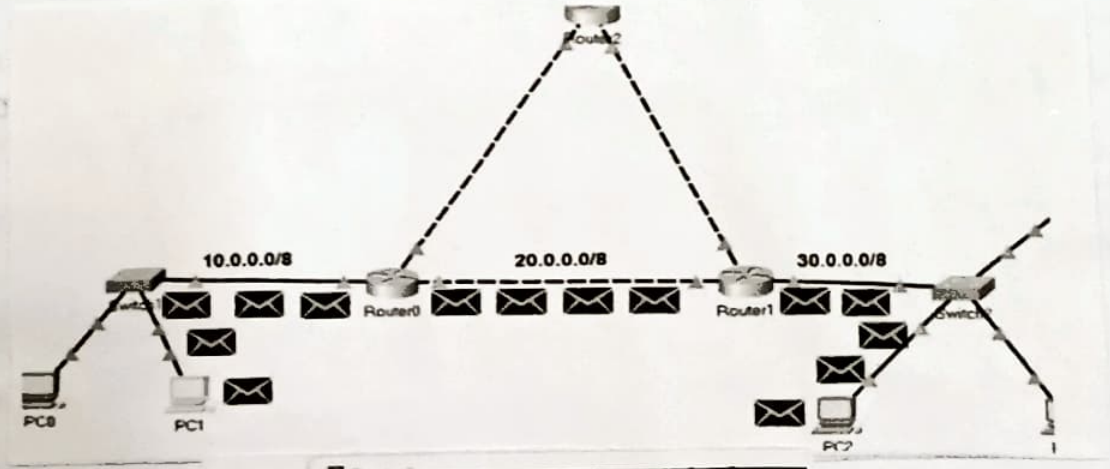
Physical Config CLI Attributes

IOS Command Line Interface

```
Router>enable
Router#configure terminal
Enter configuration commands, one per line.  End with
Router(config)#ip route 10.0.0.0 255.0.0.0 40.0.0.1
Router(config)#ip route 30.0.0.0 255.0.0.0 50.0.0.2
Router(config)#exit
Router#show ip route static
S     10.0.0.0/8 [1/0] via 40.0.0.1
S     30.0.0.0/8 [1/0] via 50.0.0.2

Router#
```

The simulate the failure of main route you can delete the link between router 0 & router 1



Command Prompt

```
Packet Tracer PC Command Line 1.0
PC>ipconfig

FastEthernet0 Connection:(default port)
Link-local IPv6 Address.....: FE80::260:70F
IP Address.....: 20.0.0.2
Subnet Mask.....: 255.0.0.0
Default Gateway.....: 20.0.0.1

PC>ping 10.0.0.2

Pinging 10.0.0.2 with 32 bytes of data:

Request timed out.
Reply from 10.0.0.2: bytes=32 time=3ms TTL=126
Reply from 10.0.0.2: bytes=32 time=3ms TTL=126
Reply from 10.0.0.2: bytes=32 time=3ms TTL=126

Ping statistics for 10.0.0.2:
    Packets: Sent = 4, Received = 3, Lost = 1
Approximate round trip times in milli-seconds:
    Minimum = 3ms, Maximum = 3ms, Average = 3ms

PC>
```

RESULT:-

This experiment has been completed successfully.

EXP 11 b) SIMULATE RIP USING CISCO PACKET TRACER

AIM:-
Simulate RIP using CISCO Packet Tracer

INITIAL IP configuration :-

Device	Interface	IP configuration	Connected with
PC0	Fast ethernet	10.0.0.2/8	Router 0's Fa0/1
Router 0	Fa 0/1	10.0.0.1/8	PC0's fast ethernet
Router 0	Se 0/0/1	192.168.1.254/30	Router 2's S/0/0 11
Router 1	Se 0/0/1	192.168.1.259/30	Router 0's S0/0/1
Router 1	Se 0/0/1	192.168.1.256/30	Router 0's S0/0/1
PC1	Fast ethernet	20.0.0.2/30	Router 2's Fa0/1.

Assign IP address to PC's

Double click PC's and click desktop menu item and click IP config

Router 0 :-

Router (config) # interface fast Ethernet 0/0

Router (config-if) # ip address 10.0.0.1

255.0.0.0 Router (config) # no shutdown.

Router (config-if) # exit.

Router 1

Router > enable

Router # configure

Router (config) # interface serial 0/0/0

Router (config-if) # no shutdown

Router (config-if) # exit

Router 2

Router > enable

Router # configure terminal

Router (config) # interface fast Ethernet 0/0

Router (config-if) # ip address 20.0.0.1 255.0.0.0 Router (config-if) # no shutdown

Router (config-if) # exit

Access the command prompt of PC, and use ping command to test connectivity from PC0

RESULT:-

... has been completed